YIBO LIN

Associate Professor ◇ Department of Design Automation and Computing System
School of Integrated Circuits, 5 Yiheyuan Road, Beijing, China, 100871 ◇ Peking University
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RESEARCH INTERESTS

Modeling and optimization in VLSI CAD, machine learning applications, and heterogeneous computing

EDUCATION

University of Texas at Austin, Texas, USA Aug. 2013 – May 2018
Ph.D., Department of Electrical and Computer Engineering

Shanghai Jiao Tong University, Shanghai, China Sep. 2009 – Jun. 2013
B.S., Department of Microelectronics

EXPERIENCE

Peking University, Beijing, China Aug. 2025 – present
Associate Professor

Peking University, Beijing, China Jul. 2019 – Jul. 2025
Assistant Professor
Department of Design Automation and Computing System,
School of Integrated Circuits (since Nov. 2021)
Center for Energy-efficient Computing and Applications (CECA),
School of Electronics Engineering and Computer Science

University of Texas at Austin, Texas, U.S. Jun. 2018 – Jun. 2019
Postdoc
Department of Electrical and Computer Engineering

TEACHING EXPERIENCE

Practice in AI-Assisted VLSI Circuit Design		
Instructor	Graduate	2026
Fundamental Algorithms for Engineering in Integrated Circuits		
Instructor	Graduate	2023–2026
Design Automation and Computing System		
Instructor	Graduate	2022–2024
Optimization and Machine Learning in VLSI Design Automation		
Instructor	Undergraduate	2021–2024
Introduction to Computing B		
Instructor	Undergraduate	2020–2022

AWARDS AND HONORS

DAC Under-40 Innovators Award	DAC	2026
Best Paper Award Nomination (13/544)	DAC	2026
Best Paper Award Nomination (×3) (12/145)	ISEDA	2026
Best Paper Award Nomination (19/266)	ICCAD	2025
Best Paper Award Nomination (7/420)	DAC	2025
Honorable Mention (×2) (/135)	ISEDA	2025
Best Paper Award Nomination (13/168)	ASPDAC	2025
Best Paper Award Nomination (10/195)	ICCAD	2024
Best Paper Award (6/151) & Honorable Mention	ISEDA	2024
Early Career Award (only one per year)	CCF Technical Committee in IC	2023
Inaugural Best Reviewer Award	ICCAD	2023
Best Paper Award (4/205)	DATE	2023
Best Paper Award (4/249)	DATE	2022
Best Paper Award Nomination (6/132)	ICCAD	2022
Donald O. Pederson Best Paper Award (2/3495 in 4 years)	TCAD	2021
Best Paper Award	ISPD	2020
Best Paper Award Nomination (12/86)	ASPDAC	2020
Best Paper Award (1/201) & Nomination (5/201)	DAC	2019
Best Paper Award Nomination	ISPD	2019
Inaugural Best Paper Award	Integration, the VLSI Journal	2018
Franco Cerrina Memorial Best Student Paper Award	SPIE	2016
A. Richard Newton Young Student Fellow	DAC	2014

PROFESSIONAL SERVICE

Executive Committee Member

- General co-chair, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD) 2025–2026
- Program co-chair, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD) 2024
- Panel chair, IEEE International Symposium of Electronics Design Automation (ISEDA) 2023–2024
- Financial chair, ACM/IEEE Workshop on Machine Learning for CAD (MLCAD) 2021–2023

Journal Editor

- Associate Editor, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024 – present
- Associate Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES), 2024 – present
- Associate Editor, Elsevier Integration, the VLSI Journal (Integration), 2024 – present
- Guest Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2024
- Guest Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2022

Contest Chair

- Contest Problem Chair, Interated Circuits EDA Elite Challenge 2021 – 2023

Technical Program Committee Member

- ACM/IEEE Design Automation Conference (DAC): 2020
- IEEE/ACM International Conference on Computer-Aided Design (ICCAD): 2018 – 2021, 2023
- Design, Automation and Test in Europe Conference (DATE): 2025
- IEEE International Conference on Computer Design (ICCD): 2019
- IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC): 2021 – 2022
- ACM International Symposium on Physical Design (ISPD): 2020, 2025
- ACM/IEEE Workshop on Machine Learning for CAD (MLCAD): 2021
- ACM Great Lakes Symposium on VLSI (GLVLSI): 2024
- Workshop on Synthesis And System Integration of Mixed Information technologies (SASIMI): 2021
- IEEE Electron Devices Technology and Manufacturing Conference (EDTM): 2021
- IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS): 2022.

Journal Reviewer

- IEEE Transaction on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- IEEE Transactions on Computers (TC)
- ACM Transaction on Design Automation of Electronic Systems (TODAES)
- SPIE Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3)
- Elsevier, Integration, the VLSI Journal (Integration)

Volunteer

- ACM SIGDA Website Administrator, 2021 – present

10 SELECTED PUBLICATIONS

Inverse chronological order. * denotes corresponding authors.

1. Yifan Chen, Zaiwen Wen, Yun Liang and **Yibo Lin***, "Stronger Mixed-Size Placement Backbone Considering Second-Order Information", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco, CA, Oct 29-31, 2023.

2. Haoyi Zhang, Xiaohan Gao, Haoyang Luo, Jiahao Song, Xiyuan Tang, Junhua Liu, **Yibo Lin***, Runsheng Wang and Ru Huang, "SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Apr 17-19, 2023. (**Best Paper Award**)

3. Jing Mai, Jiarui Wang, Zhixiong Di and **Yibo Lin***, "Multi-Electrostatic FPGA Placement Considering SLICEL-SLICEM Heterogeneity, Clock Feasibility, and Timing Optimization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2023.

4. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "Accelerating Static Timing Analysis using CPU-GPU Heterogeneous Parallelism", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2023.

5. Qipan Wang, Xiaohan Gao, **Yibo Lin***, Runsheng Wang and Ru Huang, "DeePEB: A Neural Partial Differential Equation Solver for Post Exposure Baking Simulation in Lithography", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Diego, CA, Nov 01-03, 2022. (**Best Paper Nomination**)

6. Zizheng Guo and **Yibo Lin***, "Differentiable-Timing-Driven Global Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.

7. Siting Liu, Peiyu Liao, Zhitang Chen, Wenlong Lv, **Yibo Lin*** and Bei Yu*, "FastGR: Global Routing on CPU-GPU with Heterogeneous Task Graph Scheduler", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022. (**Best Paper Award**)

8. Zizheng Guo, Jing Mai and **Yibo Lin***, "Ultrafast CPU/GPU Kernels for Density Accumulation in Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.

9. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.

10. **Yibo Lin***, Zixuan Jiang, Jiaqi Gu, Wuxi Li, Shounak Dhar, Haoxing Ren, Brucek Khailany and David Z. Pan, "DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2021. (**Best Paper Award**)

PUBLICATIONS

Book Chapters

[B3] **Yibo Lin**, Zizheng Guo and Jing Mai, "Deep Learning Framework for Placement", Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)

[B2] Haoyu Yang, **Yibo Lin** and Bei Yu, "Machine Learning for Mask Synthesis and Verification", Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)

Conference and Journal Papers (* denotes corresponding authors)

Summary: DAC (37), ICCAD (32), IEEE TCAD (34), DATE (21), etc.

[J251] Zhenkun Lin, **Yibo Lin**, Genggeng Liu* and Gang Du, "URoute: Universal Routability Prediction", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J250] Xu He*, Yushan Wang, Chenjing Yang, Renjun Zhao, **Yibo Lin**, Peiyu Liao, Bei Yu, Yao Wang, Chang Liu and Yang Guo, "Efficient Timing Prediction and Optimization Using Derivable Gradient Boosting Machine Model at Placement Stage", ACM Transactions on Design Automation of Electronic Systems (TODAES), 2026. (accepted)

[J249] Chunyuan Zhao, Zizheng Guo, Zuodong Zhang and **Yibo Lin***, "GAP-LA: GPU-Accelerated Performance-Driven Layer Assignment", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J248] Xufeng Yao, Jiaxi Jiang, Yuxuan Zhao, Peiyu Liao, **Yibo Lin** and Bei Yu*, "EvoPlace: Evolution of Optimization Algorithms for Global Placement via Large Language Models", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J247] Tianxiang Zhu, Qipan Wang, **Yibo Lin*** and Runsheng Wang*, "M2MOR: Multi-scale Thermo-mechanical Coupling Simulation of Chiplet Heterogeneous Integration Systems Based on Model Order Reduction", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J246] Haoyi Zhang, Jiechen Huang, Bingyang Liu, Wenjian Yu*, **Yibo Lin*** and Runsheng Wang, "APEC-Route: Guiding Analog Routing with On-the-Fly Parasitic Extraction Based on a Neural Capacitance Model", ACM Transactions on Design Automation of Electronic Systems (TODAES), 2026. (accepted)

[C245] Mengjia Dai, Hongduo Liu, Yuhao Ji, Xiaojing Su, **Yibo Lin** and Bei Yu*, "Soft-Constrained Triple Patterning Layout Decomposition", IEEE International Conference on Computer Design (ICCD), Hong Kong, China, Nov 16-18, 2026. (accepted)

[C244] Xiang Li, Zuodong Zhang and **Yibo Lin***, "TAT: Timing-Aware Steiner Tree Generation for Buffer Insertion", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C243] Yuxi Lin, **Yibo Lin**, Zepeng Li, Weijie Fang, Guangyong Chen and Genggeng Liu*, "Clock Tree Synthesis with Cross-Tier Skew and Wirelength Co-Optimization for Face-to-Face Hybrid Bonding 3D ICs", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C242] Zhenkun Lin, Taoyu Zhou, Qipan Wang, Tianxiang Zhu, **Yibo Lin**, Meng Li, Runsheng Wang, Yandong He and Gang Du*, "VULCAN: Unified Multi-Scale Chip Thermal Simulation System for Design-Technology Co-Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C241] Jingxin Wang, Sinian Zhang, Yun Liang, **Yibo Lin**, Pengpeng Ren, Runsheng Wang and Weikang Qian*, "GUALO: A Generalizable Uncertainty-Aware AI Agent for Logic Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C240] Jincheng Lou, Ruohan Xu, Jiecheng Ma, Boyang Wu, Yuhao Ji, Chunyuan Zhao, Xinyu Qu, Yuxiang Zhao, Runzhe Tao, Bei Yu, Zhixiong Di and **Yibo Lin***, "LEGO 2.0: A Unified Agentic Platform for Digital Chip Design and Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C239] Weijian Fan, Zhiyuan Zhao, Jinpeng Wang, Haoyi Zhang, Jincheng Lou, Runsheng Wang, Xiyuan Tang and **Yibo Lin***, "VinePlace: Compact Analog and Mixed-Signal Placement via Core-Centered Vine Representation", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C238] Yuhao Ji, Siying Yu, Yunqi Shi, Fangzhou Liu, Yuxuan Zhao, Jiaxi Jiang, Zizheng Guo,

Chao Qian, **Yibo Lin** and Bei Yu*, "HeteroOpt3D: Timing Optimization for Heterogeneous Hybrid-bonding-based Face-to-Face 3D ICs", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C237] Kairong Guo, Haoyi Zhang, Haichuan Liu, Chunyuan Zhao and **Yibo Lin***, "ScoutCell: Sensitivity-Guided Coupling Optimization for Timing-Driven Standard Cell Synthesis", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C236] Zixuan Pan, Zizheng Guo, Yufan Du, Runsheng Wang and **Yibo Lin***, "Disentangled Differentiable Timing-Power Co-Optimization with Quad-Gradient Gate sizing", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (**Best Paper Nomination**)

[C235] Yi Ren, Yukun Wang, Xiang Meng, Guoyao Cheng, Baokang Peng, Lining Zhang, **Yibo Lin**, Runsheng Wang and Guangyu Sun*, "CellE: Automated Standard Cell Library Extension via Equality Saturation", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C234] Jing Mai, Yi-Chen Lu, **Yibo Lin** and Haoxing Ren, "LATTE: Legality-Assured Differentiable Timing-Driven Detailed Placement", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C233] Wenbo An, Kairong Guo, Haoyi Zhang and **Yibo Lin***, "EPiCell: Electro-Physical Co-Modeling for Standard Cell PPA Prediction", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C232] Yuhao Ji, Yunqi Shi, Tianshu Hou, Yuxuan Zhao, Chunyuan Zhao, Peiyu Liao, Zizheng Guo, Chao Qian, **Yibo Lin** and Bei Yu*, "Integrated Timing-driven Placement for Hybrid-Bonding-based Face-to-Face 3D ICs", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C231] Kexing Zhou, Youwei Zhuo, **Yibo Lin**, Weikang Qian, Pengpeng Ren and Yun Liang*, "Activity-Aware Partitioning for Effective Multi-Threaded Event-Driven RTL Simulation", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C230] Jiechen Huang, Hector R. Rodriguez, Dingcheng Yang, Zuochang Ye, **Yibo Lin** and Wenjian Yu*, "AttentionCap: Transformer Based Capacitance Matrix Learning Toward Full-Chip Extraction", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C229] Haoyi Zhang, Weijian Fan, Xiaohan Gao, Bingyang Liu, Runsheng Wang and **Yibo Lin***, "PANDA: An LLM-Enhanced Performance-Driven Analog Design Framework Bridging Design Intent and Layout Generation", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (**Invited Paper**)

[C228] Haoyi Zhang, Kairong Guo, Bojie Zhang, **Yibo Lin*** and Runsheng Wang, "FusionCell: Cross-Attentive Fusion of Layout Geometry and Netlist Topology for Standard-Cell Performance Prediction", International Conference on Machine Learning (ICML), Seoul, Korea, Jul 6-11, 2026.

[C227] Xizhe Shi, Zizheng Guo, Zuodong Zhang, Yun Liang, Runsheng Wang* and **Yibo Lin***, "HeteroPower: A CPU-GPU Heterogeneous Engine to Accelerate Gate-Level Power Analysis",

IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026. (**Best Paper Nomination**)

[C226] Kairong Guo, Jiechen Huang, Wenjian Yu and **Yibo Lin***, "CAPCell: Standard Cell Layout Synthesis with Parasitic Capacitance Aware Parallel Sampling", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C225] Jincheng Lou, Ruohan Xu, Jiecheng Ma, Runzhe Tao, Xinyu Qu and **Yibo Lin***, "LEGO: An LLM Skill-Based Front-End Design Generation Platform", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026. (**Best Paper Nomination**)

[C224] Zizhuo Fu, Yifan Zhou, Zhaoxin Lu, Guangyu Sun, Runsheng Wang, Meng Li* and **Yibo Lin***, "RePart: Efficient Hypergraph Partitioning with Logic Replication Optimization for Multi-FPGA System", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C223] Xun Jiang, Haoran Lu, Yifan Chen, Jiarui Wang, Chunyuan Zhao, Yibo Zhang, Jianxiang Jin, Heng Wu, Runsheng Wang and **Yibo Lin***, "Flexible Double-Side Pin Redistribution with Efficient Virtual-Net-Guided Cell Type Assignment", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026. (**Best Paper Nomination**)

[C222] Haichuan Liu, Zizheng Guo, Runsheng Wang and **Yibo Lin***, "ElmoreCeff: A GPU-Friendly Elmore-Like Delay Calculator with a Closed-Form Effective Capacitance Model", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C221] Bingyang Liu, Haoning Jiang, Haoyi Zhang, Xiaohan Gao, Zichen Kong, Xiyuan Tang, David Z. Pan and **Yibo Lin***, "GRAIN: A Design-Intent-Driven Analog Layout Migration Framework", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C220] Yuxiang Zhao, Jing Mai, Zishu Li, Haoyi Zhang, Jincheng Lou, Qing He, Runsheng Wang and **Yibo Lin***, "PowerCube: A Formula-in-the-Loop Framework for Cross-Stage Power Map Prediction", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C219] Mingjun Wang, Yihan Wen, Yuntao Lu, Fengrui Liu, Yuxiang Zhao, Boyu Han, Jianan Mu, **Yibo Lin**, Runsheng Wang, Huawei Li and Bei Yu, "CircuitNet 3.0: A Multi-Modal Dataset with Task-Oriented Augmentation for AI-Driven Circuit Design", International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, Apr 23-27, 2026.

[C218] Chenglin Ye, Yuze Lu, Yizhan Liu, Ligong Zhang, Jinghan Xu, Fei Liu, **Yibo Lin**, Zheng Zhou and Xiaoyan Liu, "EMAPER: Cross-Level Electromigration Aware Placement and Routing EDA Workflow for Interconnects Hotspot Prediction and Mitigation", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C217] Jiajun Tan, Qichao Ma, Yiming Du, Yiming Gan, Ling Liang, **Yibo Lin**, Ming Zhu, Zongwei Wang and Yimao Cai, "SONIC: Smart Optimization for Neural-Integrated CMP with Timing-Aware Fills", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C216] Zizheng Guo, Runsheng Wang and **Yibo Lin***, "Provably Optimal Planar Pareto Nearest Neighbor Search with Double Monotone Chains", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C215] Jianqing Chen, Zhenkun Lin, Xun Jiang, Genggeng Liu*, **Yibo Lin** and Gang Du, "DSR: A Systematic Approach for Efficient Double-Sided Signal Routing", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C214] Siting Liu, Peng Xu, Peiyu Liao, Keren Zhu, **Yibo Lin** and Bei Yu*, "Submodular Maximization-Inspired Adaptive Routing Bend Space Planning", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C213] Tianxiang Zhu, Qipan Wang, **Yibo Lin*** and Runsheng Wang*, "Efficient Warpage Simulation Of Complex 2.5-D/3-D IC Structures with Novel Meshing Algorithm and Layerwise Plate Theory", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C212] Yuhao Ji, Zizheng Guo, **Yibo Lin** and Bei Yu*, "HeteroSTA3D: A CPU-GPU Heterogeneous Static Timing Analysis Engine for 3D ICs", China Semiconductor Technology International Conference (CSTIC), Shanghai, Mar, 2026. (**Invited Paper**)

[C211] Jiarui Wang, Runzhe Tao, Jing Mai, Xun Jiang, Shenghua Wang, Cuiliu Yang, Haoyu Jie, Kan Huang, Richard Y. Sun and **Yibo Lin***, "TDM Signal Grouping and Package Pin Assignment for 2.5D Multi-FPGA Systems with Lookahead Placement", International Symposium on Field-Programmable Gate Arrays (FPGA), Monterey, CA, Feb 22-24, 2026.

[C210] Zizheng Guo, Haichuan Liu, Xizhe Shi, Shenglu Hua, Zuodong Zhang, Chunyuan Zhao, Runsheng Wang and **Yibo Lin***, "HeteroSTA: A CPU-GPU Heterogeneous Static Timing Analysis Engine with Holistic Industrial Design Support", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026. (**Invited Paper**)

[C209] Xizhe Shi, Zizheng Guo, **Yibo Lin***, Zuodong Zhang, Yun Liang and Runsheng Wang*, "HeteroLatch: A CPU-GPU Heterogeneous Latch-Aware Timing Analysis Engine", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[C208] Kairong Guo, Haoran Lu, Rui Guo, Jiarui Wang, Chunyuan Zhao, Heng Wu, Runsheng Wang and **Yibo Lin***, "Standard Cell Layout Synthesis for Dual-Sided 3D-Stacked Transistors", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[C207] Weijian Fan, Haoyi Zhang, Weibin Lin, Runsheng Wang and **Yibo Lin***, "MOSTAR: Multi-Stage Hierarchical Bayesian Optimization for Substructure-Aware High-Dimensional Analog Circuit Sizing", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[C206] Chenhao Xue, Kezhi Li, Jiaxing Zhang, Yi Ren, Zhengyuan Shi, Chen Zhang, **Yibo Lin**, Lining Zhang, Qiang Xu and Guangyu Sun*, "AC-Refiner: Efficient Arithmetic Circuit Optimization Using Conditional Diffusion Models", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[C205] Jingchao Hu, **Yibo Lin**, Hao Yu, Quan Chen, Zhou Jin* and Cheng Zhuo, "IR Drop-Aware ECO: A Fast Approach to Minimize Layout and Timing Disturbance", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[J204] Siting Liu, Ziyi Wang, Fangzhou Liu, **Yibo Lin**, Bei Yu* and Martin Wong, "Sign-off Timing Considerations via Concurrent Routing Topology Optimization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[J203] Zhenkun Lin, Genggeng Liu*, Xing Huang, **Yibo Lin**, Jixin Zhang, Wenhao Liu and Ting-Chi Wang, "A Unified Deep Reinforcement Learning Approach for Constructing Rectilinear and Octilinear Steiner Minimum Tree", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[J202] Jingchen Zhu, Chenhao Xue, Yiqi Chen, Zhao Wang, Chen Zhang, Yu Shen, Yifan Chen, Zekang Cheng, Yu Jiang, Tianqi Wang, **Yibo Lin**, Wei Hu, Bin Cui, Runsheng Wang, Yun Liang and Guangyu Sun*, "Theseus: Exploring Efficient Wafer-Scale Chip Design for Large Language Models", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[J201] Yajuan Su, Zixi Liu, **Yibo Lin**, Xiaojing Su, Yuqin Wang, Xin Hong, Yujie Jiang, Pengyu Ren and Yayi Wei, "A Post-Routing Layout Optimization Framework for Lithography Process Window Enlargement", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[J200] Jing Mai, Chunyuan Zhao, Zuodong Zhang, Zhixiong Di, Runsheng Wang and **Yibo Lin***, "LEGALM 2.0: A Versatile Augmented Lagrangian Method-Based Methodology for Mixed-Cell-Height Legalization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[C199] Chunyuan Zhao, Jiarui Wang, Xun Jiang, Jincheng Lou and **Yibo Lin***, "GTA: GPU-Accelerated Track Assignment with Lightweight Lookup Table for Conflict Detection", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025. (**Best Paper Nomination**)

[C198] Yufan Du, Zizheng Guo, Runsheng Wang and **Yibo Lin***, "Differentiable Physical Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C197] Yuhao Ji, Yuntao Lu, Zuodong Zhang, Zizheng Guo, **Yibo Lin** and Bei Yu*, "DiffCCD: Differentiable Concurrent Clock and Data Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C196] Yuan Pu, Yuhao Ji, Siying Yu, Zuodong Zhang, Zizheng Guo, Zhuolun He, **Yibo Lin**, David Z. Pan and Bei Yu*, "GPU Acceleration for Versatile Buffer Insertion", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C195] Yi Ren, Baokang Peng, Chenhao Xue, Kairong Guo, Yukun Wang, Guoyao Cheng, **Yibo Lin**, Lining Zhang and Guangyu Sun*, "Orthrus: Dual-Loop Automated Framework for System-Technology Co-Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C194] Haichuan Liu, Zizheng Guo, Runsheng Wang and **Yibo Lin***, "IncreGPUSTA: GPU-Accelerated Incremental Static Timing Analysis for Iterative Design Flows", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C193] Tianxiang Zhu, Qipan Wang, **Yibo Lin*** and Runsheng Wang*, "High-Resolution Full-Chip Thermal Resistance Extraction of BEOL Interconnects in 3-D ICs Considering Detailed Via Connectivity", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C192] Chenchen Zhao, Zhengyuan Shi, Xiangyu Wen, Chengjie Liu, Yi Liu, Yunhao Zhou, Yuxiang

Zhao, Hefei Feng, Yinan Zhu, Waa Gwok-Wan, Xin Cheng, Weiyu Chen, Yongqi Fu, Chujie Chen, Chenhao Xue, Ying Wang, **Yibo Lin**, Jun Yang, Ning Xu, Xi Wang and Qiang Xu*, "MMCircuitEval: A Comprehensive Multimodal Circuit-Focused Benchmark for Evaluating LLMs", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[J191] Bingyang Liu, Haoyi Zhang, Xiaohan Gao, Zichen Kong, Xiyuan Tang, **Yibo Lin***, Runsheng Wang and Ru Huang, "LayoutCopilot: An LLM-powered Multi-agent Collaborative Framework for Interactive Analog Layout Design", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Aug, 2025.

[C190] Yuxiang Zhao, Zhuomin Chai, Xun Jiang, Qiang Xu, Runsheng Wang and **Yibo Lin***, "DeepLayout: Learning Neural Representations of Circuit Placement Layout", International Conference on Machine Learning (ICML), Vancouver, Canada, Jul 13-19, 2025.

[C189] Lijie Zeng, Jiatai Sun, Xiao Wu, Dan Niu, Tianshi Wang, **Yibo Lin**, Zuochang Ye and Zhou Jin*, "G-SpNN: GPU-Accelerated Passivity Enforcement for S-Parameter Modeling with Neural Networks", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.

[C188] Ziyang Yu, Peng Xu, Zixiao Wang, Binwu Zhu, Qipan Wang, **Yibo Lin**, Runsheng Wang, Bei Yu* and Martin Wong, "SDM-PEB: Spatial-Depthwise Mamba for Enhanced Post-Exposure Bake Simulation", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.

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